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ORIGINAL RESEARCH

NBDNet: A Deep Learning Algorithm for Despeckling of SAR Data

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Abstract
In the recent years, remote sensing has gained high momentum in varied applications. Satellite imaging and processing is one of the most sorted techniques followed by researchers. Synthetic aperture radar (SAR) images are popular among the remote sensing community due to its capability of imaging in all weather conditions. The practical applications of SAR data is limited due to presence of speckle noise. In the past, deep learning methods are developed to despeckle the SAR images. This paper proposes a convolutional neural network based non-blind denoising network (NBDNet) for the despeckling of SAR images. In the proposed NBDNet model, attention blocks are introduced to preserve the structural and texture details of the captured scene globally. Further, squeeze and excitation module and convolutional block attention module have been used in the proposed NBDNet to capture the minute structural information of the artefacts. The experimental results of proposed NBDNet and benchmark algorithms are evaluated on synthetic UC merced land-use images and real SAR images. Quantitative and visual results of of proposed NBDNet yield better texture and structural detail preservation as compared to benchmark algorithms on both datasets.

Keywords Remote sensing · SAR processing · Speckle noise · Convolutional neural network

Introduction

Synthetic aperture radar imaging (SAR) has gained huge attention in the remote sensing community due to its potential of being invariant to different weather and geographical conditions. However, SAR images are contaminated with speckle noise impairing further downstream processing of images such as segmentation, change detection. The constructive and destructive interference of reflected radar signal contributes to speckle noise. As a result, SAR image despeckling performance has a significant impact on a variety of remote sensing processes. Smoothing is done to reduce the granularity of the content. Smoothing affects edge preservation [7]. Effectiveness of the despeckling method can be gauged either by visual inspection or by performance metrics. Reduction in granular content, clarity in the artifacts, preservation of edges in heterogeneous areas, clear differentiation of low contrast objects, smoothness of homogeneous patches and texture preservation are some of the aspects to be stressed during visual inspection.

Since 1980s scholars have developed different despeckling methodologies that can be broadly categorised as non-deep learning and deep learning based. SAR data is often represented as a multiplicative model. It is the product of terrain backscatter with a Gaussian distribution and speckle noise with a Gamma distribution. The Gamma distribution of speckle noise is characterised by one parameter known as the number of looks (L). The SAR image becomes more noisy as the value of L decreases. Non-deep learning approaches use a multi-look processing strategy [9]. In SAR imaging, another key factor to consider is the incidence angle. The incidence angle in SAR imaging refers to the angle formed by the radar wave and the normal at the target. Unwanted distortions in SAR images include layover and foreshortening due to terrain slopes smaller than or larger than the incidence angle, respectively. This makes identifying topographic features challenging. As a result, SAR despeckling becomes difficult [27]. Non-deep learning methods include spatial filters and transform domain filters. Spatial adaptive filter based despeckling approach are developed such as Lee filter [15], Kuan filter [13], Frost filter [10]. In spatial domain approaches, pixel based noise

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filtering is performed. Major shortcomings of the spatial based despeckling approaches is the loss of texture details and over smoothing in the heterogeneous area. Further, filters were designed exploiting the characteristic features of transform domain. Meer filter [18], rational Laplacian pyramid filter [1] and homomorphic filters [2, 26] in wavelet transform domain are some prominent work in transform domain. Transform domain filters showcased decent quantitative results along with preservation of structural details compared to spatial domain filters. Parrilli et al. [23] suggested a block matching 3D-based SAR despeckling algorithm (SAR-BM3D). The block-matching stage employs a probabilistic similarity metric, and the wavelet shrinkage is created by employing an additive signal-dependent noise model and seeking the best local linear minimum-mean-square-error estimator in the wavelet domain.

In the last one decade, deep learning approaches are being extensively adapted by remote sensing community. Convolutional neural networks (CNN) are frequently employed in the development of SAR despeckling deep learning models. Supervised and unsupervised training are the two strategies that are used for training deep learning models. Unsupervised training has the advantage of not requiring the ground truth, compared to supervised training. SAR image despeckling faces the challenge of unavailability of noiseless ground truth. So researchers have acclimatized the use of synthetic images for training the deep learning models. Speckle noise is formulated as multiplicative noise [24], is employed by many researchers in generating synthetic SAR dataset from optical color images. Another noteworthy approach is the use of multi-look SAR image as reference image in supervised training. Difference in imaging technique of optical images and SAR images is yet another challenging aspect for the use of synthetic dataset in deep learning models.

Wang et al. [30] proposed a CNN model to train the network using a residual division layer to estimate the speckle. Integrating pixel-to-pixel euclidean loss, perceptual loss, and adversarial loss, Wang et al. [31] created a generative adversarial network-based despeckling model. A 17-layer convolutional neural network (CNN) based despeckling architecture using a multitemporal technique was designed by Chiercia et al. [4]. This model suppressed the spatial features by overly smoothing homogeneous and heterogeneous areas. Yue et al. [35] trained a convolution neural network to learn the probability distribution of the radar cross section. By making the two-scatterer assumption, real SAR images are used in training the model. Zhang et al. [36] designed a dilated residual network (SAR-DRN) employing the L2 loss for training the network. Low-level features are propagated through the network by interleaving dilated convolutions and the skip connections every two layers. However, they encountered significant challenges when trying to despeckle the incredibly heterogeneous locations. Lattari et al. [14] pre-trained a encoder–decoder based U-Net model with synthetic dataset and fine tuned it using real SAR images. This method produced sharper reconstructions that were less prone to the formation of artefacts and blurring effects. Li et al. [16] designed a hybrid dilated convolutional framework and attention module with skip connections. Ma et al. [17] designed a coupled convolutional neural network trained with a pair of noisy images having same texture but different noise realisation. Molini et al. [19] employed hybrid training with both synthetic and real SAR image to train their blind spot convolutional neural network. Smoothing across homogeneous areas only reaches a modest degree in blind spot CNN model. Cai et al. [3] addressed the shortcoming of edge distortion by using split images with extended boundary. This approach enhanced the attention towards the boundary pixels. Dalsasso et al. [5] introduced U-Net based architecture with semi-supervision. Firstly, network is trained with synthetically produced speckled images. Then, images are randomly selected from a time-series in pairs. The second image is being corrected for alterations using reflectivities estimated using the network trained on a synthetic dataset. Vitale et al. [29] suggested a CNN model that is trained using a multi-objective cost function to find statistical patterns, strong scatterers, and spatial features in the noisy image. Single channel speckled SAR image is transformed into multi-channel features employing dilated convolution based self calibrated block to extract contextual features. Subsequently, multi-channel features are reconstructed into single channel despeckled image in the model proposed by Yuan et al. [33]. Vitale et al. [28] proposed yet another CNN model by using image generated by multilook real SAR images as reference. It was possible to strike a balance between the preservation of spatial characteristics in homogeneous areas and urban structures with this approach. A self-supervised bayesian despeckling model was developed by Molini et al. [20], in which the speckle is described by the prior distribution of the pixels in the reference image, conditioned on their neighbouring pixels. They succeeded in creating crisp edges and lifelike details in man-made structures. Mulissa et al. [22] developed a siamese architecture and trained using temporally averaged real SAR image as reference label. This approach bypassed the prior modelling of noise. Ko et al. [12] developed a encoder-decoder network with continuous attention module and trained using a combination of modified despeckling gain and total variation loss function. This model acquires multiscale data while emphasising on the more noticeable channel-specific features. Dalsasso et al. [6] proposed a noise to noise self supervised training methodology by making use of the SAR image's real and imaginary portions. Here, the reflectivity is estimated by training the network using real part of the SAR image and loss computation with the corresponding imaginary part of SAR image. Yuan et al. [34] created a framework for adversarial learning to produce speckled to speckled image pairings from the same scene. Here, the necessity to have speckle-free SAR

image from same scene is avoided. Farhadini et al. [8] proposed convolutional neural network based architecture exploiting the features in wavelet domain. Details coefficient and approximation coefficients are estimated with fast and flexible denoising network (FFDNet) and averaged maximum a posteriori (AMAP) estimator. Despeckled image is constructed by applying inverse transform to the combination of details coefficient and approximation coefficients.

The key contributions of this manuscript are as follows:

- We propose a robust deep learning model NBDNet for the despeckling of SAR images.
- We conduct extensive experiments and demonstrated that the proposed NBDNet performs better as compared to benchmark algorithms and models both on the synthetic SAR images and real-world SAR images.

The rest of this manuscript is formulated as follows: "Proposed Model" focuses on the detailed mathematical analysis of the proposed NBDNet model. "Training and Implementation Details" focuses on the details about training and implementation of the proposed NBDNet network and quality metrics used for analysis of experimental results. In "Results and Discussion", the experimental results of proposed model and benchmark algorithms are discussed. The conclusion of the manuscript is presented in "Conclusion".

Proposed Model

The proposed NBDNet model is designed for the despeckling of SAR images. The model is trained using the speckle noise described by Eq. (1):

$$\rho_n(n) = \frac{L^L n^{L-1} exp(-nl)}{\gamma(L)} \tag{1}$$

in the above equation, the mathematical function represented by the symbol γ, known as the gamma function, widens a factorial calculation to non-integer values. The equivalent number of glances is denoted by the symbol L.

Deep learning attention network enables to focus only on the important parts of the input data. By allowing the model to selectively emphasize or suppress certain elements, their importance or relevance can be captured for the task at hand. When interpreting various inputs for machine learning models, the attention mechanism comes into play by determining relevance through intelligent weighing assignments. By assigning greater weightings for crucial inputs and lesser scores for those that hold little importance, critical inputs can be made immediately available with non-critical aspects being deemphasized. This proves most useful for managing complex structures and extended interdependency relationships that are characteristic of many advanced models as attention promotes adaptability through its dynamic prioritizing of important inputs.

The architecture of the proposed NBDNet model is shown in Fig. 1. It consists of the noise estimation subnetwork and non blind denoising subnetwork. Further, non blind denoising network has squeeze and excitation attention module, convolutional block attention module and convolutional layers with skip connections [12]. The noise estimation subnetwork's objective is to calculate the statistical characteristics of the noise that is present in the input signal. For proper denoising or signal restoration, this information is essential. The succeeding processing steps of the model may better adapt to and eliminate the noise by modeling and predicting the noise characteristics, which leads to higher output quality. The non-blind denoising subnetwork takes a noisy input image and attempts to create a denoised output image by successfully suppressing the noise while maintaining crucial image information. Typically, it is trained using pairs of noisy and clean images, with the clean images serving as the denoising task's ground truth. By adjusting the channel-wise feature responses adaptively, the squeeze and excitation attention module is a method that improves the representational strength of convolutional neural networks (CNNs). It focuses on understanding channel interdependence and gives various channels varying weights for relevance. Also, the convolutional block attention has two parts i.e. continuous attention module (CAM) and the spatial attention module (SAM).

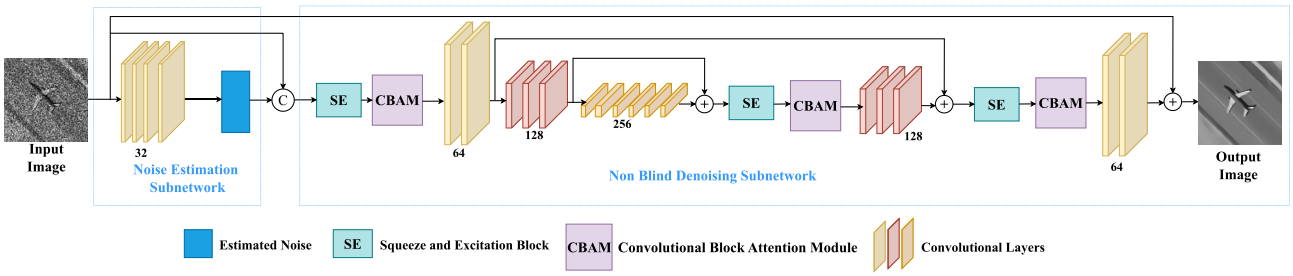


Fig. 1 Proposed NBDNet model for despeckling

Once trained, the non-blind denoising network can remove noise from new, unseen images by passing them through the network. The network leverages its learned knowledge of noise characteristics to denoise the input images and reconstruct clean versions and passes this output to the attention modules. The description of the attention modules is discussed in the subsequent sections.

Squeeze and Excitation Attention Module (SE)

To improve channel-wise feature recalibration or attention in deep learning models, the SE was put together. It allows the network to selectively highlight informative channels while squashing less important ones, thereby capturing fine-grained information and improving feature representation. Figure 2 represents the block diagram of SE. Its simplicity, efficiency, and effectiveness make the SE attention module a popular choice for enhancing the representational power of deep learning models, enabling them to capture more discriminative features and achieve state-of-the-art performance.

The SE attention module consists of two major steps: squeeze and excitation. In the squeeze step, a global pooling operation (typically average pooling) is applied to the input feature map's spatial dimensions, bringing down the spatial dimensions to $1 \times 1 \times C$ while preserving the channel-wise information. This pooling operation aims to capture the global context of the feature map. Following the squeeze step, the excitation step utilizes fully connected (FC) layers to model the channel-wise dependencies and relationships between the channels. These FC layers, often accompanied by non-linear activation functions like ReLU, learn to capture the interdependencies among the channels and generate channel-wise activation weights.

To ensure that the activation weights are within a meaningful range, a sigmoid activation function is applied to rescale them between 0 and 1. This rescaling step allows the network to emphasize important channels while suppressing less relevant ones, creating an attention mechanism. Finally, the original feature map is element-wise multiplied with the calibrated channel-wise activation weights to create a feature map that emphasises informative elements while attenuating less significant ones.

Convolutional Block Attention Module (CBAM)

CBAM is a widely used attention mechanism for the denoising, classification and segmentation tasks. By explicitly modelling the interdependencies between spatial and channel-wise features, it improves the representational capability of CNNs. It has two main attention modules i.e. continuous attention module (CAM) and spatial attention module (SAM) [12]. The architecture of CBAM is shown in Fig. 3.

The channel attention module is shown in the Fig. 4. It comprises of max pooling and average pooling. The max pool layer helps in targeting the detailed information. The distinct features were captured for the fine detailing. Likewise average pooling computes the statistical properties. So, here both average and max pooling techniques are used simultaneously. The effectiveness of using both at a time is high than using them seperately. F^{c}_{max} and F^{c}_{avg} denotes the max pooling and average pooling features respectively. The shared MLP layers, which comprise the multi layer perceptron, receive both of these features. This layer helps in extracting the important data while reducing the dimensions. The dimension before passing to the MLP is $M_c \in R^{1\times1\times C}$ and after the MLP layer the dimension have been reduced to $M_c \in R^{1\times1\times C/r}$, here r indicates the reduction ratio. The overall channel attention is computed as given in Eq. (2).

$$\mathbf{M_c}(\mathbf{F}) = \sigma\left(MLP\left(\mathbf{F^c_{max}}\right) + MLP\left(\mathbf{F^c_{avg}}\right)\right) \tag{2}$$

where σ is the sigmoid activation function

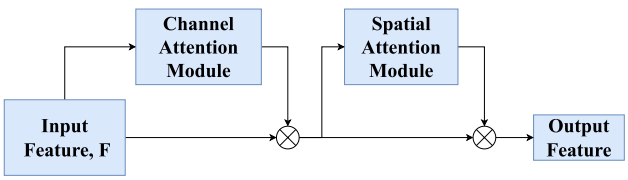


Fig. 3 Convolutional block attention module [12]

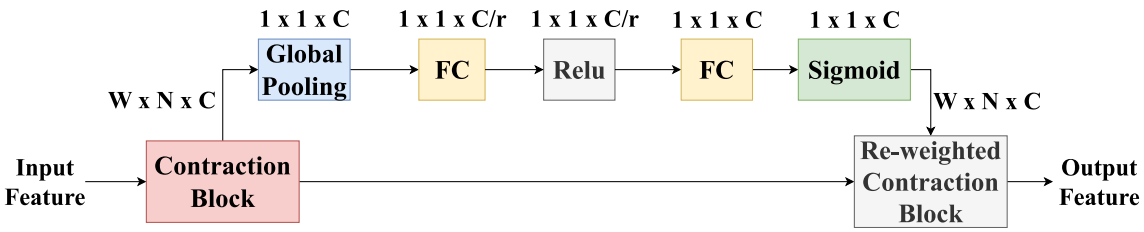


Fig. 2 Squeeze and excitation attention module [11]

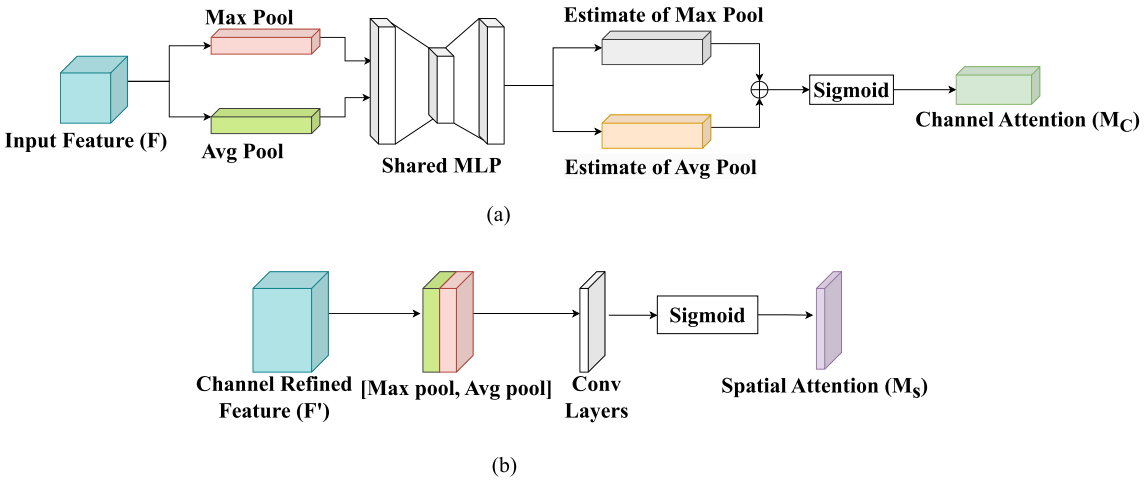


Fig. 4 **a** Channel attention module, **b** spatial attention module [12]

The output of the channel attention module, or the channel refined feature multiplied with F, is the input of the spatial attention module. Spatial attention module focuses on the spatially significant features. As illustrated in the Fig. 4, two 2D maps are created by combining channel information from a feature map using the max pooling and average pooling operations along the channel axis, they are $M^s_{avg} \in R^{H\times W\times 1}$ and $M^s_{max} \in R^{H\times W\times 1}$. The 2D spatial attention map is created by convolving the concatenated 2D maps with a standard convolution layer. It is evaluated as in Eq. (3):

$$\mathbf{M_s}(\mathbf{F'}) = \sigma\left(f^{3\times 3}\left(\left[\mathbf{M^s_{max}}(\mathbf{F'}); \mathbf{M^s_{avg}}(\mathbf{F'})\right]\right)\right) \tag{3}$$

here σ indicates the sigmoid activation function.

Training and Implementation Details

Dataset and Preprocessing

To train the model, a dataset containing diverse images is necessary. UC merced land use dataset [32], which encompasses 21 distinct classes with 100 images in each class is used for generating synthetic SAR images. For the training purpose out of each class 70 images are taken and made into 64×64 patches. Totally 22,900 patches are made and converted to grayscale and normalized using min-max normalization approach. Out of 22,900 patches, 21,000 patches are used as training set and the remaining 1900 patches are for validation set. Noise levels of L = 1, 2, 4 were added to these patches. For the testing purpose 3 images are considered from the remaining images of the UC merced land use dataset. Napoli image is considered outside the dataset. For the real SAR data, Sentinel-1 and TerraSAR images are tested with the trained model.

Training Setup

This model was run on the Google Colab using Python 3.6 and Tensorflow. For rapid and successful training, the Adam optimizer with a constant learning rate of 0.001 was chosen. All the models are trained for 20 epochs with batch size of 2. For model training, mean square error (MSE) loss between clean and noisy image pairs is adopted.

Quality Metrics

The performance of despeckling techniques on simulated and real SAR images is evaluated with and without reference quality measures. Peak signal to noise ratio (PSNR) [25], Structural Similarity Index (SSIM) [25], root mean square error (RMSE) [21], correlation coefficient (CC) [21] and mean bias error (MBE) [21] are the with-reference metrics computed to evaluate the performance of synthetic SAR test images. Equivalent number of looks (ENL) [25], mean of image (MOI) [25] are the without-reference indices used to evaluate real SAR test images.

Results and Discussion

The proposed NBDNet architecture quality metrics and despeckled images are compared with other existing despeckling methodologies like SAR-BM3D [23], IDCNN [30], SAR-DRN [36], U-Net [14] and SAR-CAM [12]. The discussion about despeckling performance on synthetic images and real SAR images are presented in "Synthetic

SAR Data Experiment" and "Real SAR Data Experiment" respectively.

Synthetic SAR Data Experiment

PSNR and SSIM are the quality metrics calculated to assess the despeckling performance on synthetic SAR images. The PSNR values calculated for Building, Airplane, Intersection and Napoli images corresponding to different looks (L = 1, 2, 4) speckle noise are listed in Table 1 and the best PSNR values are highlighted with bold font. Higher the PSNR value, better is the speckle removal. Table 1 implies that, in comparison to all other state-of-the-art despeckling models taken into account, the proposed model achieves higher PSNR values. Better noise filtering and structure retention are indicated by SSIM values that are nearer to 1. As highlighted in Table 1, SSIM values corresponding to proposed model are more closer to 1 for all the test images except for Napoli image with L = 4 speckle noise.

Visual comparison of the despeckled test images is captured in Figs. 5, 6 and 7. The building's edges and structural integrity are better preserved in the despeckled image of the proposed model, as seen in Fig. 5. However, in homogeneous areas like the roof top, there is over smoothing. The airplane's despeckled image generated by proposed model as shown in Fig. 6 clearly shows the runway markings, but the texture of the land around the runway is somewhat obscured. The intersection image of the proposed model, as shown in Fig. 7, also exhibits a similar despeckling pattern, however the texture features are slightly less visually pleasing. RMSE, CC and MBE values are computed between the ground truth and the corresponding despeckled SAR images for Building, Airplane and Intersection. Lower the RMSE value, better is the despeckling performance. The bar-plot of the RMSE values is shown in Fig. 8. It compares the RMSE values of NBDNet with other deep learning models considered for comparison. Bar-plots indicate lower RMSE value for NBDNet despeckled SAR images in comparison to all other models under consideration for

Table 1 PSNR and SSIM values for synthetic images with different looks

Image	Models	PSNR			SSIM		
		L = 1	L = 2	L = 4	L = 1	L = 2	L = 4
Building	Noisy	7.5	10.42	13.44	0.08	0.13	0.22
	SAR-BM3D [23]	21.94	23.97	26.12	0.61	0.72	0.78
	IDCNN [30]	22.58	25.01	27.17	0.65	0.79	0.85
	SAR-DRN [36]	23.59	25.26	26.91	0.77	0.81	0.85
	U-Net [14]	23.77	25.20	27.11	0.76	0.81	0.85
	SAR-CAM [12]	23.42	25.21	27.41	0.62	0.69	0.75
	NBDNet	**24.30**	**26.56**	**28.63**	**0.78**	**0.83**	**0.86**
Airplane	Noisy	6.49	9.39	12.45	0.05	0.09	0.15
	SAR-BM3D [23]	23.12	24.12	25.65	0.59	0.71	0.75
	IDCNN [30]	22.74	25.00	26.37	0.62	0.77	0.83
	SAR-DRN [36]	24.01	24.89	26.18	0.75	0.78	0.83
	U-Net [14]	24.35	24.96	26.44	0.76	0.79	0.83
	SAR-CAM [12]	24.22	25.18	26.19	0.61	0.72	0.76
	NBDNet	**24.84**	**26.03**	**27.02**	**0.78**	**0.81**	**0.84**
Intersection	Noisy	5.72	8.65	11.75	0.07	0.12	0.20
	SAR-BM3D [23]	21.54	22.64	24.61	0.60	0.73	0.76
	IDCNN [30]	21.54	23.87	25.47	0.62	0.78	0.83
	SAR-DRN [36]	22.38	23.80	25.23	0.72	0.78	0.82
	U-Net [14]	22.62	23.71	25.29	0.76	0.79	0.81
	SAR-CAM [12]	22.41	23.87	25.41	0.65	0.74	0.76
	NBDNet	**23.34**	**24.70**	**26.40**	**0.77**	**0.80**	**0.84**
Napoli	Noisy	7.15	10.03	13.17	0.11	0.17	0.28
	SAR-BM3D [23]	20.19	21.99	23.67	0.42	0.52	0.61
	IDCNN [30]	20.77	23.05	24.51	0.64	0.76	0.81
	SAR-DRN [36]	21.21	23.05	24.33	0.71	0.77	**0.85**
	U-Net [14]	21.02	22.96	24.41	0.71	0.76	0.81
	SAR-CAM [12]	21.32	23.07	24.67	0.67	0.73	0.78
	NBDNet	**22.48**	**23.87**	**24.98**	**0.72**	**0.77**	0.82

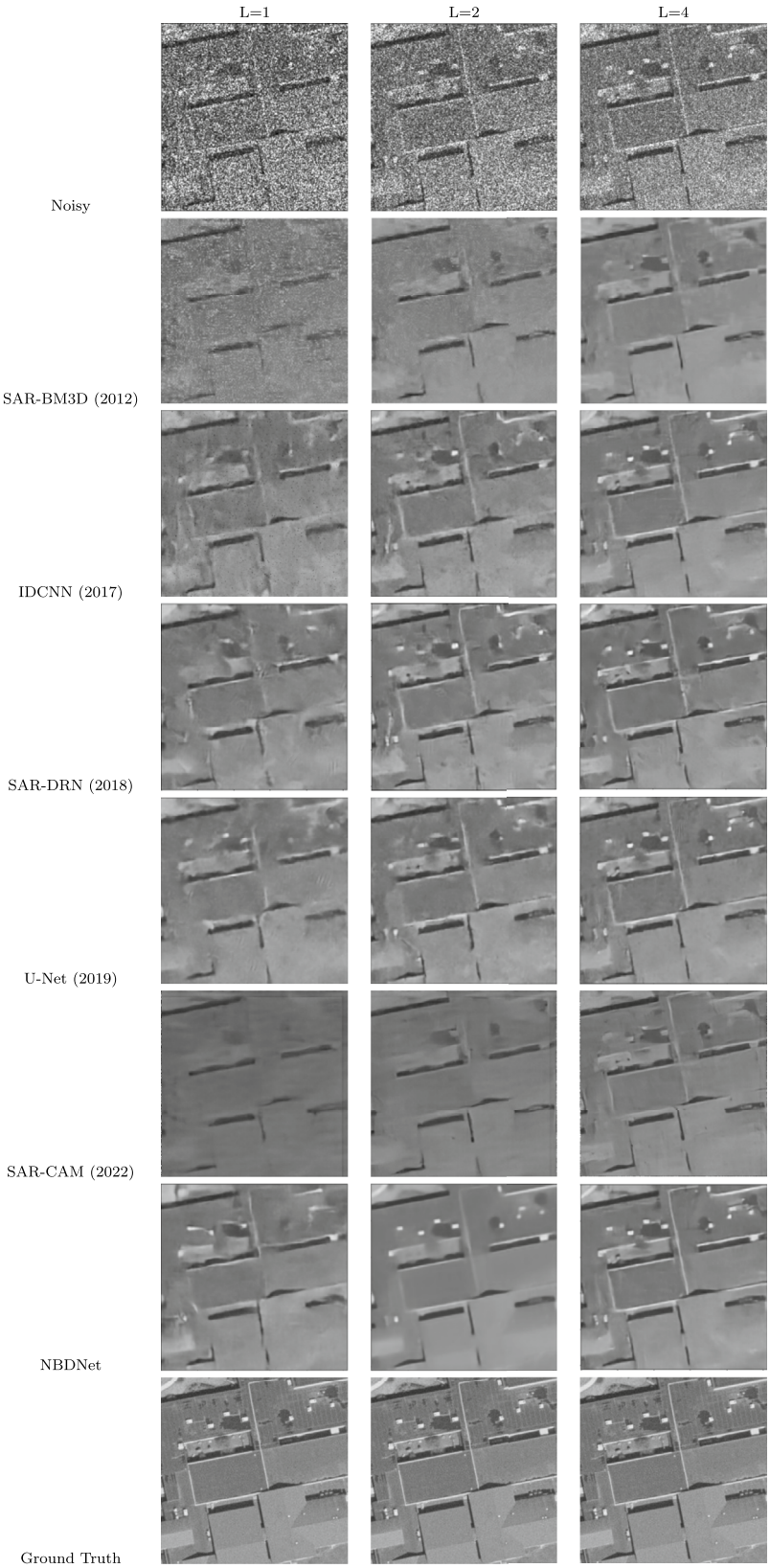


Fig. 5 Prediction results for Building image for L = 1, 2, 4 speckle noise

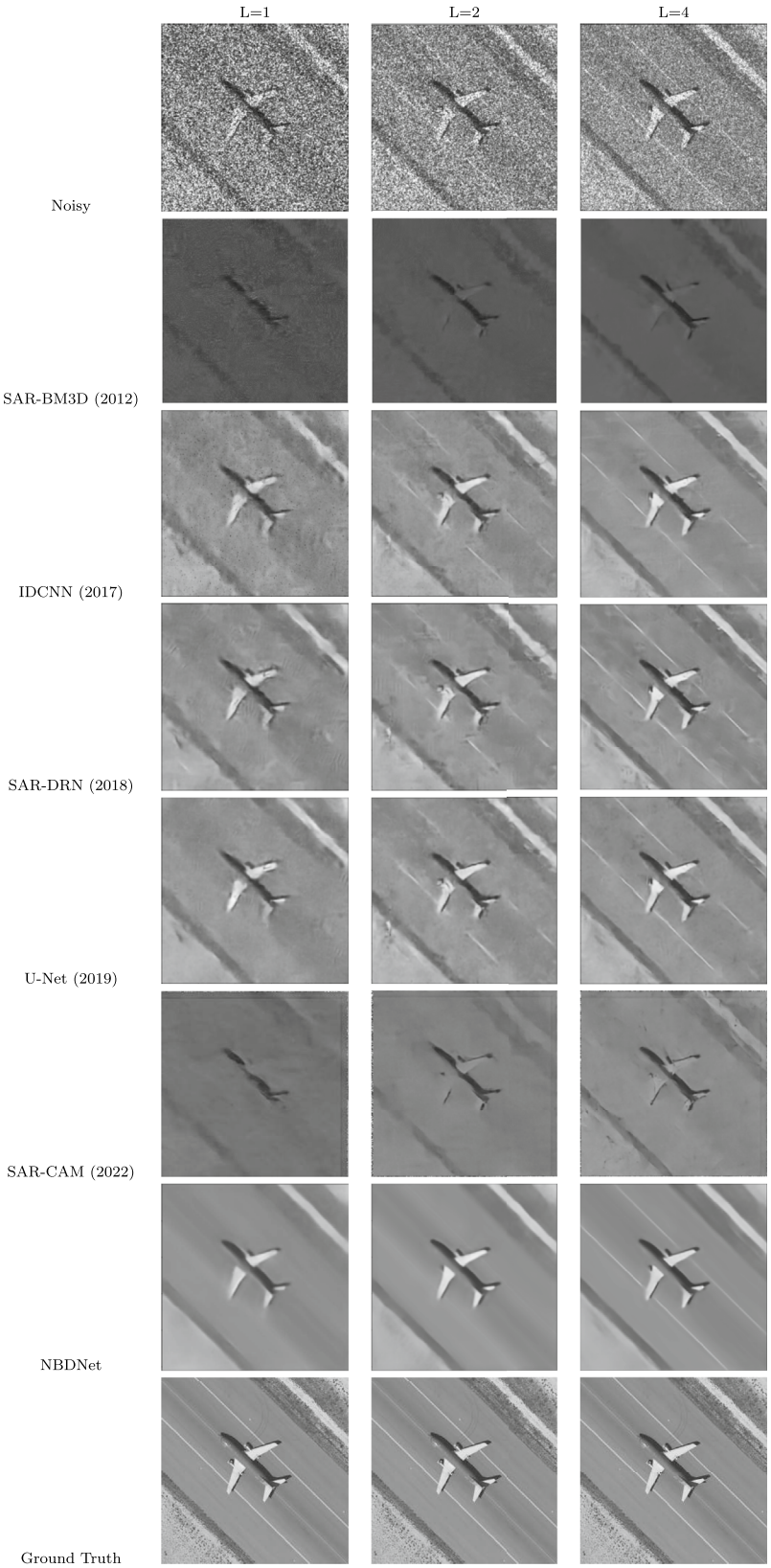


Fig. 6 Prediction results for airplane image for L = 1, 2, 4 speckle noise

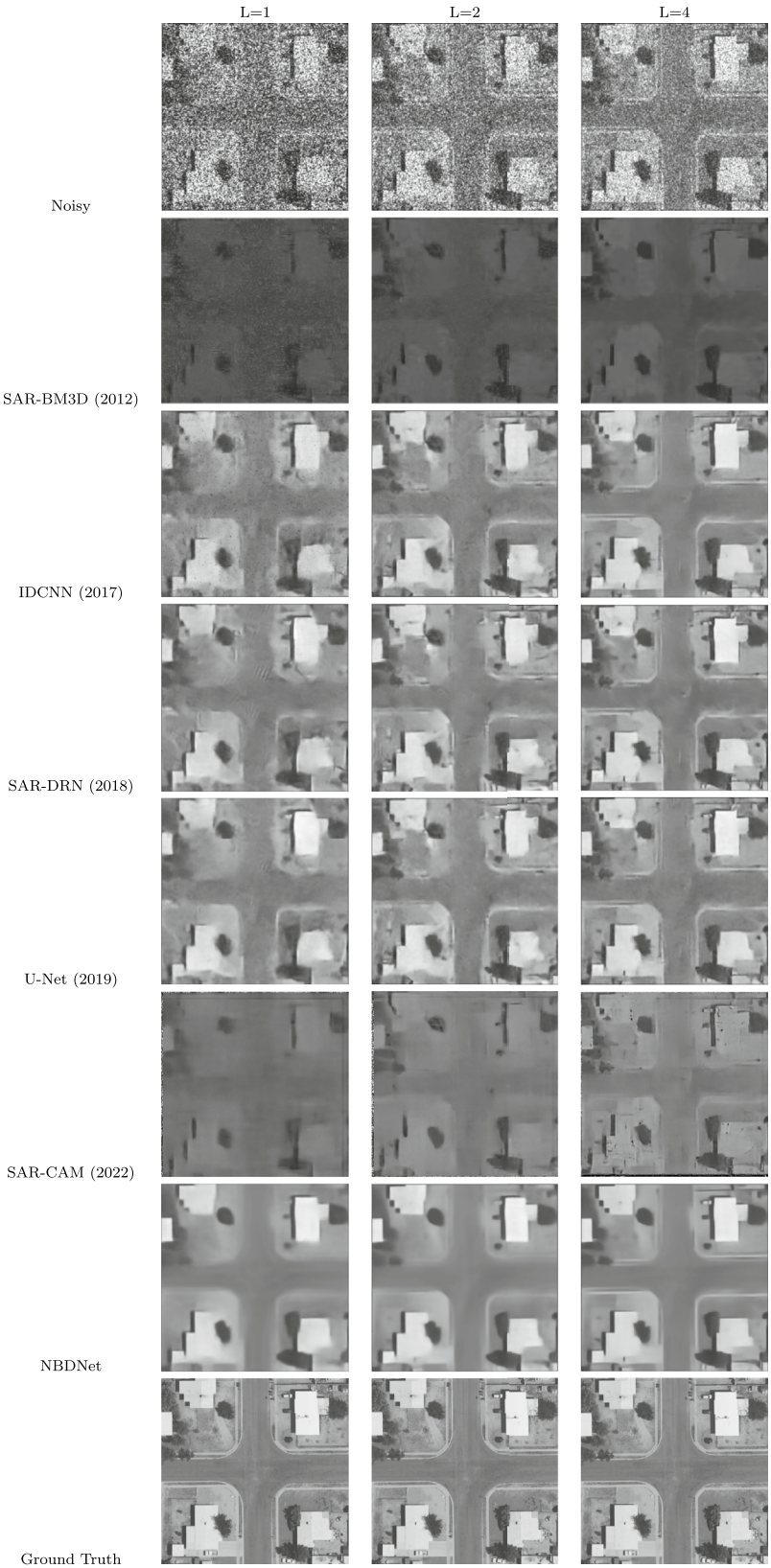


Fig. 7 Prediction results for intersection image for L = 1, 2, 4 speckle noise

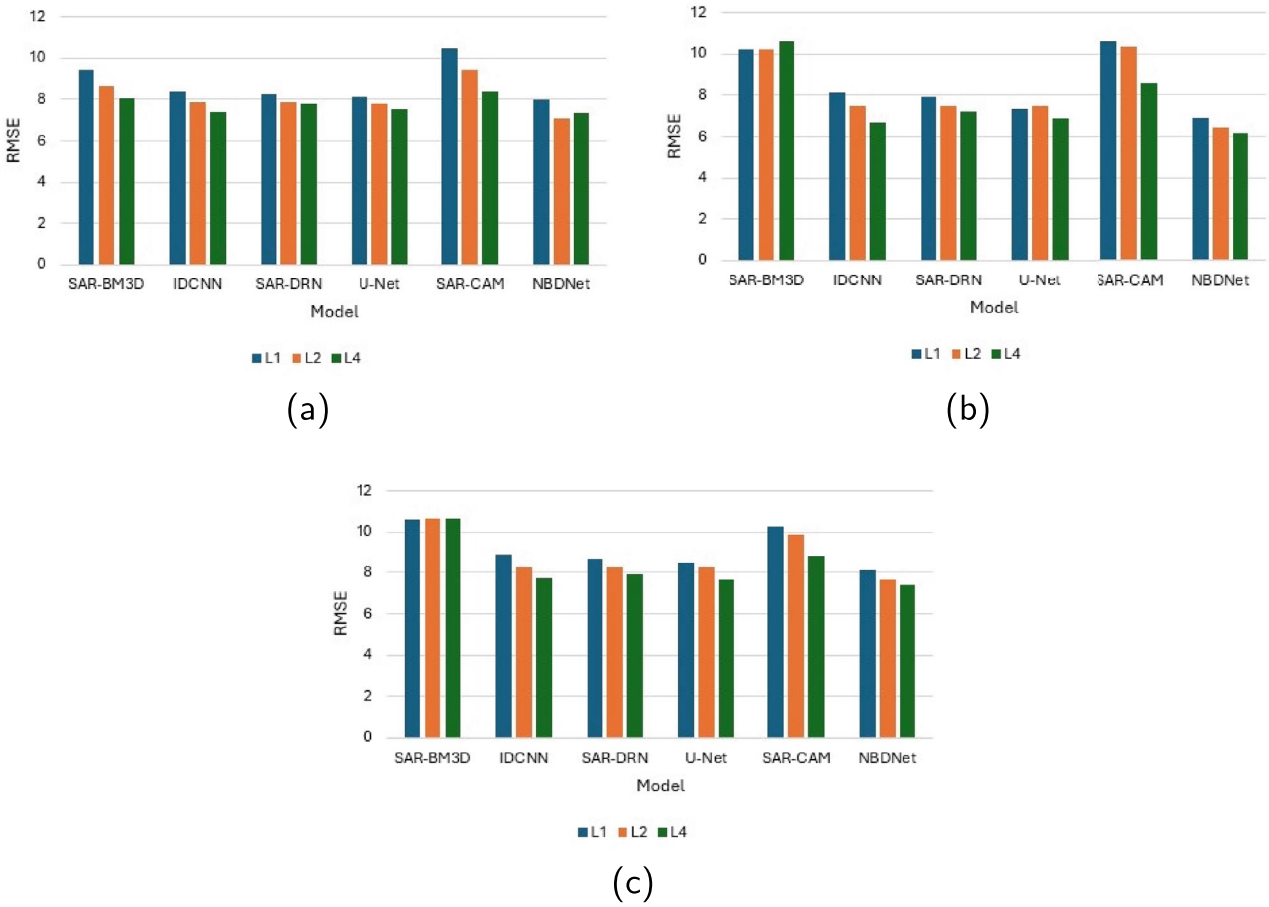


Fig. 8 **a** RMSE for Building, **b** RMSE for Airplane, **c** RMSE for Intersection dataset

L = 1, 2, 4 speckle noise levels. The closer the despeckled SAR image is to the ground truth, the higher the CC value. The bar-plot of the CC values is shown in Fig. 9. CC value for NBDNet model despeckled SAR images is higher than all other models under consideration except for Building L = 1, 4 despeckled SAR images. The closer the MBE value is to 0, the lesser the mean deviation of individual pixels in the despeckled image from the ground truth. The bar-plot of the CC values is shown in Fig. 10. NBDNet's MBE is not the lowest among the models evaluated for the majority of speckle levels across Building, Airplane and Intersection datasets. This can be attributed to NBDNet's inappropriate smoothing effect. This causes poor texture preservation, as seen in visual comparisons.

Real SAR Data Experiment

On two real SAR images from Sentinel-1 and TerraSAR, the performance of the proposed method and recent state-of-the-art methods is compared. In order to evaluate the effectiveness of various image despeckling techniques, ENL and MoI are estimated from the homogeneous area outlined in red in noisy images of Fig. 11. The corresponding results are also tabulated in Table 2 and the best values are highlighted in bold font. Higher the ENL value better is the despeckling. Compared to the other comparison models, the proposed model produced improved ENL values for both test images. MoI value is expected to be closer to 1 in the homogeneous region. MoI evaluated for the proposed model is comparable to MoI of other models under consideration. Visual comparison of real SAR test images is shown in Fig. 11. The NBDNet demonstrated aesthetically pleasing artefact and texture preservation.

Conclusion

This paper presented NBDNet model for despeckling of SAR data. A good despeckled SAR image should have reduced granularity, improved contrast, smoothing as well as texture preservation in homogeneous regions, and edge preservation in non-homogeneous regions. PSNR, SSIM,

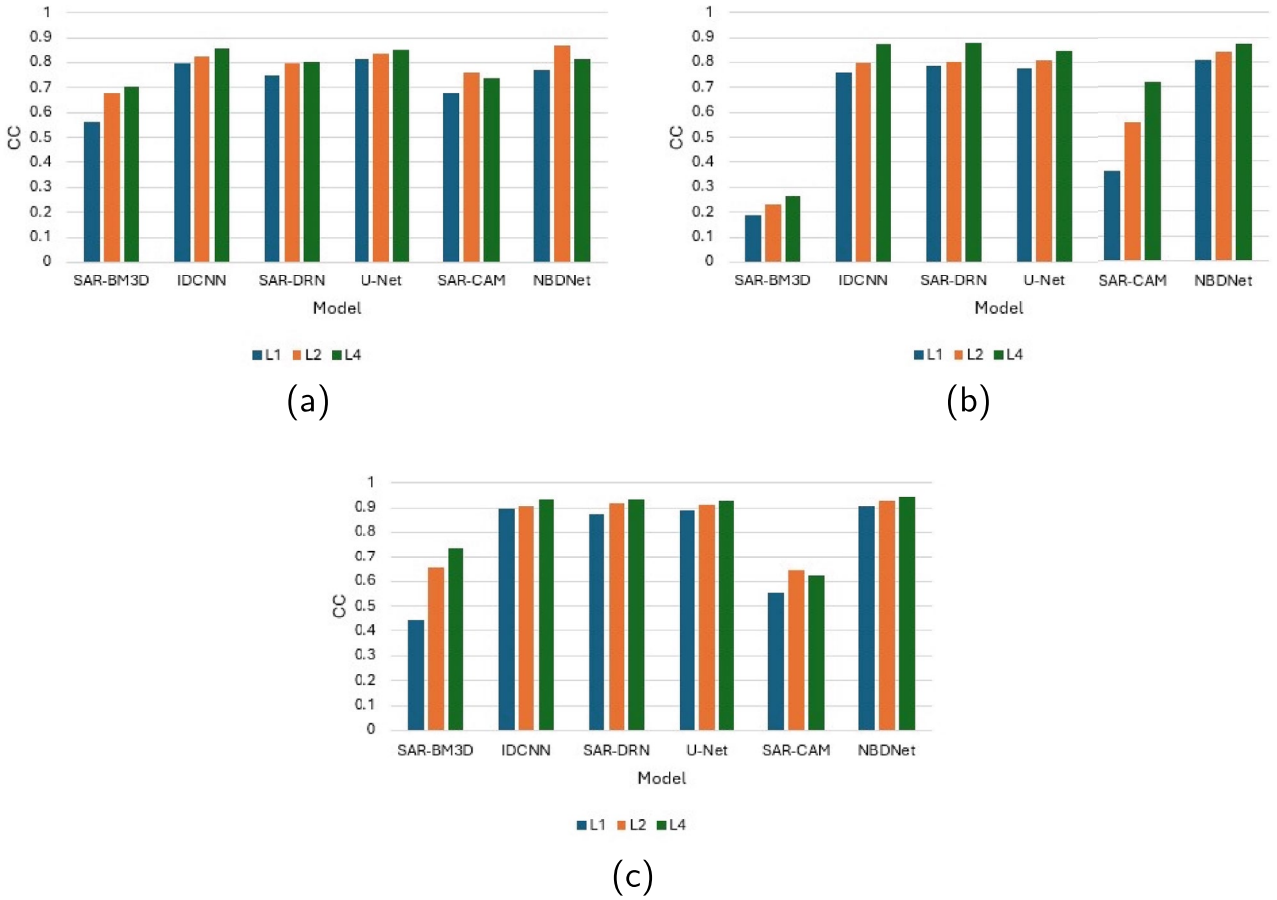


Fig. 9 **a** CC for Building, **b** CC for Airplane, **c** CC for Intersection dataset

RMSE, CC, MBE are the metrics computed to evaluate the performance of despeckling models on synthetic SAR data. Proposed NBDNet model resulted in better quantitative results with respect to PSNR (0.8 dB), SSIM (1%) on an average for the despeckling of synthetic SAR data with L = 1, 2, 4 speckle noise, in comparison to benchmark algorithms and models under consideration. The NBDNet model showed good smoothing in homogeneous regions, but texture information was lost, as evidenced by the CC and MBE metric values. When tested on Sentinel-1 and TerraSAR images, NBDNet performed better than other models under consideration in terms of speckle reduction, smoothing, and edge preservation. It is also reflected in a larger ENL metric value and a MOI closer to one. The distinguishing performance of the proposed NBDNet model is the impact of effective utilization of attention modules and the skip connections. Furthermore, there is scope to improve the texture and edge preservation by including feature extraction in frequency domain.

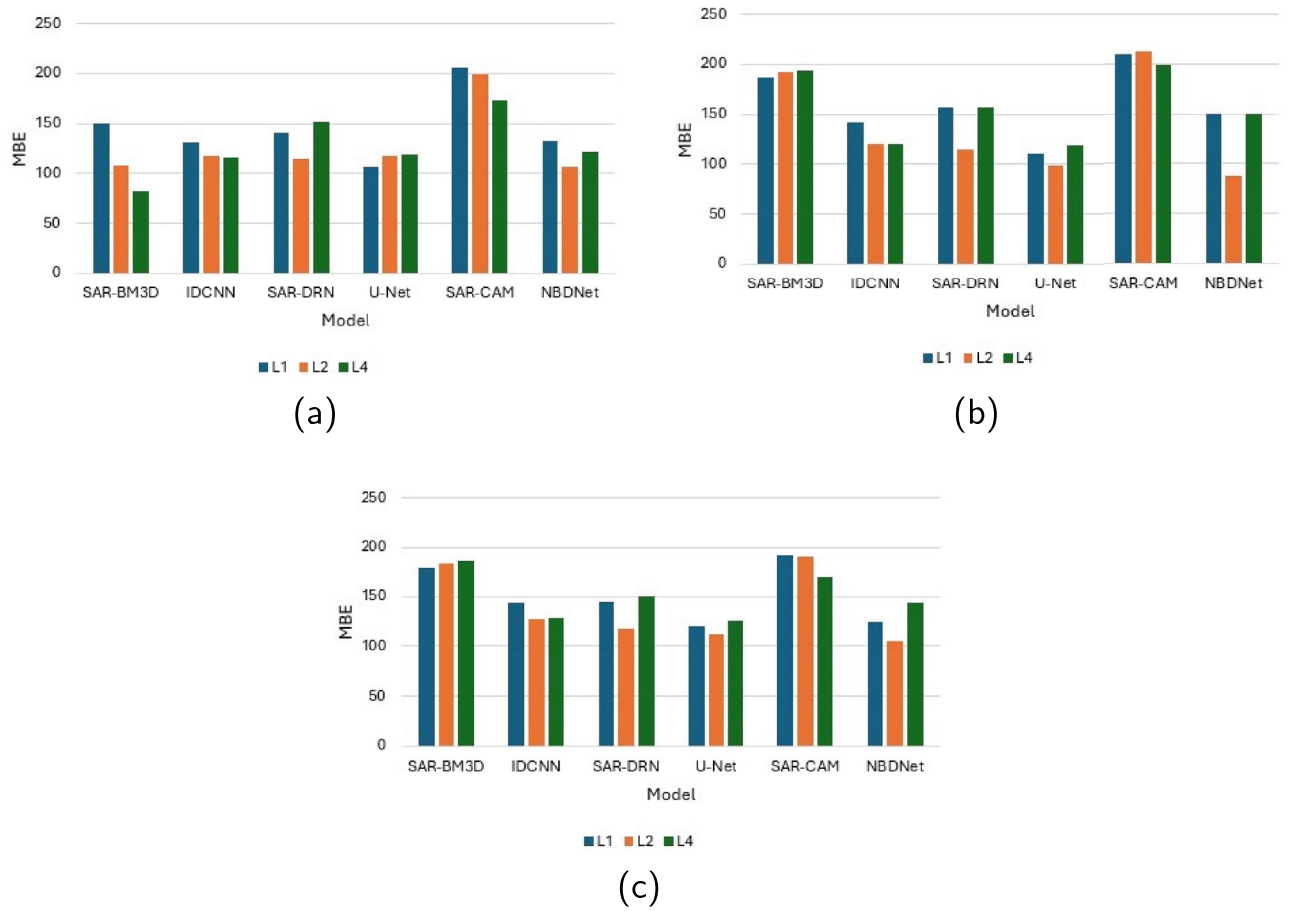


Fig. 10 **a** MBE for Building, **b** MBE for Airplane, **c** MBE for Intersection dataset

Table 2 Metrics for real SAR images

Image	Model	ENL	MOI
Sentinel-1	SAR-BM3D [23]	195.12	0.95
	IDCNN [30]	245.63	0.98
	SAR-DRN [36]	320.12	1.08
	U-Net [14]	349.74	0.98
	SAR-CAM [12]	239.63	0.98
	NBDNet	**351.61**	**1.009**
Terra-SAR	SAR-BM3D [23]	279.12	0.94
	IDCNN [30]	330.16	**1.004**
	SAR-DRN [36]	479.52	1.09
	U-Net [14]	318.02	1.06
	SAR-CAM [12]	367.68	0.99
	NBDNet	**490.85**	1.07

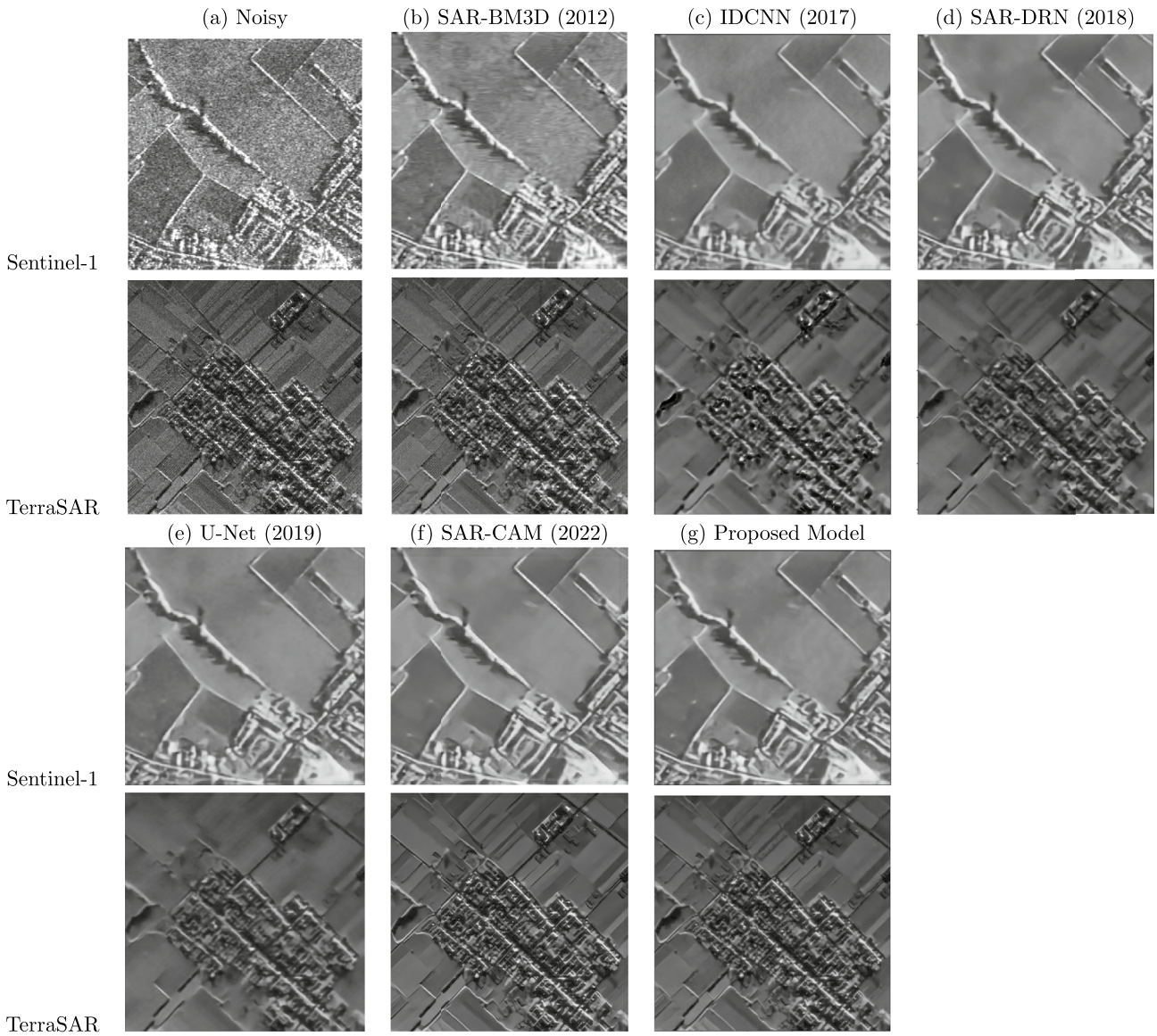


Fig. 11 Prediction results for TerraSAR and Sentinel-1 images

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Code Availability The code will be made available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors report no conflict of interest.

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References

1. Aiazzi B, Alparone L, Baronti S, et al. Pyramid-based multiresolution adaptive filters for additive and multiplicative image noise. IEEE Trans Circuits Syst II Analog Digit Signal Process. 1998;45:1092–7. https://doi.org/10.1109/82.718817.
2. Bhuiyan MI, Ahmad MO, Swamy MN. A new homomorphic Bayesian wavelet-based MMAE filter for despeckling SAR images. In: Proceedings—IEEE International Symposium on Circuits and Systems, 2005, p. 4935–8. https://doi.org/10.1109/ISCAS.2005.1465740.
3. Cai Y, Sumantyo JTS. Despeckling of synthetic aperture radar image using deep-learning model. In: Proceedings—2021 7th Asia-Pacific Conference on Synthetic Aperture Radar, APSAR 2021, 2021. https://doi.org/10.1109/APSAR52370.2021.9688432.
4. Chierchia G, Cozzolino D, Poggi G, et al. SAR image despeckling through convolutional neural networks. In: International Geoscience and Remote Sensing Symposium (IGARSS) 2017, 2017-July, p. 5438–41. https://doi.org/10.1109/IGARSS.2017.8128234.
5. Dalsasso E, Denis L, Tupin F. SAR2SAR: a semi-supervised despeckling algorithm for SAR images. IEEE J Sel Top Appl Earth Observ Remote Sens. 2021;14:4321–9. https://doi.org/10.1109/JSTARS.2021.3071864.
6. Dalsasso E, Denis L, Tupin F. As if by magic: self-supervised training of deep despeckling networks with merlin. IEEE Trans Geosci Remote Sens. 2022. https://doi.org/10.1109/TGRS.2021.3128621.
7. De Borba AA, Muhuri A, Marengoni M, et al. Feature selection for edge detection in PolSAR images. Remote Sens. 2023;15(9). https://doi.org/10.3390/rs15092479. https://www.mdpi.com/2072-4292/15/9/2479.
8. Farhadiani R, Homayouni S, Bhattacharya A, et al. SAR despeckling based on CNN and Bayesian estimator in complex wavelet domain. IEEE Geosci Remote Sens Lett. 2022. https://doi.org/10.1109/LGRS.2022.3185557.
9. Frery AC, Freitas C, Sant'anna SJ, et al. Statistical properties of SAR data and their consequences. In: Selected Papers on Remote Sensing, Space Science and Information Technology. Seminars of the United Nations Programme on Space Applications, vol 10. United Nations, New York; 1999. p. 53.
10. Frost VS, Stiles JA, Shanmugan KS, et al. A model for radar images and its application to adaptive digital filtering of multiplicative noise. IEEE Trans Pattern Anal Mach Intell PAMI. 1982;4:157–66. https://doi.org/10.1109/TPAMI.1982.4767223.
11. Hu J, Shen L, Sun G. Squeeze-and-excitation networks. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, p. 7132–41. https://doi.org/10.1109/CVPR.2018.00745.
12. Ko J, Lee S. Sar image despeckling using continuous attention module. IEEE J Sel Top Appl Earth Observ Remote Sens. 2022;15:3–19. https://doi.org/10.1109/JSTARS.2021.3132027.
13. Kuan DT, Sawchuk AA, Strand TC, et al. Adaptive noise smoothing filter for images with signal-dependent noise. IEEE Trans Pattern Anal Mach Intell PAMI. 1985;7:165–77. https://doi.org/10.1109/TPAMI.1985.4767641.
14. Lattari F, Leon BG, Asaro F, et al. Deep learning for SAR image despeckling. Remote Sens. 2019. https://doi.org/10.3390/RS11131532.
15. Lee JS. Speckle analysis and smoothing of synthetic aperture radar images. Comput Graph Image Process. 1981;17:24–32. https://doi.org/10.1016/S0146-664X(81)80005-6.
16. Li J, Li Y, Xiao Y, et al. Hdranet: Hybrid dilated residual attention network for SAR image despeckling. Remote Sens. 2019. https://doi.org/10.3390/RS11242921.
17. Ma X, Wang C, Yin Z, et al. Sar image despeckling by noisy reference-based deep learning method. IEEE Trans Geosci Remote Sens. 2020;58:8807–18. https://doi.org/10.1109/TGRS.2020.2990978.
18. Meer P, Park RH, Cho KJ. Multiresolution adaptive image smoothing. CVGIP Graph Models Image Process. 1994;56:140–8. https://doi.org/10.1006/CGIP.1994.1013.
19. Molini AB, Valsesia D, Fracastoro G, et al. Towards deep unsupervised SAR despeckling with blind-spot convolutional neural networks. In: International Geoscience and Remote Sensing Symposium (IGARSS), 2020, p. 2507–2510. https://doi.org/10.1109/IGARSS39084.2020.9324183.
20. Molini AB, Valsesia D, Fracastoro G, et al. Speckle2Void: deep self-supervised SAR despeckling with blind-spot convolutional neural networks. IEEE Trans Geosci Remote Sens. 2022. https://doi.org/10.1109/TGRS.2021.3065461.
21. Muhuri A, Gascoin S, Menzel L, et al. Performance assessment of optical satellite-based operational snow cover monitoring algorithms in forested landscapes. IEEE J Sel Top Appl Earth Observ Remote Sens. 2021;14:7159–78. https://doi.org/10.1109/JSTARS.2021.3089655.
22. Mullissa AG, Persello C, Reiche J. Despeckling polarimetric SAR data using a multistream complex-valued fully convolutional network. IEEE Geosci Remote Sens Lett. 2022. https://doi.org/10.1109/LGRS.2021.3066311.
23. Parrilli S, Poderico M, Angelino CV, et al. A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage. IEEE Trans Geosci Remote Sens. 2012;50:606–16. https://doi.org/10.1109/TGRS.2011.2161586.
24. Ranjani JJ, Thiruvengadam SJ. Dual-tree complex wavelet transform based SAR despeckling using interscale dependence. IEEE Trans Geosci Remote Sens. 2010;48:2723–31. https://doi.org/10.1109/TGRS.2010.2041241.
25. Singh P, Diwakar M, Shankar A, et al. A review on SAR image and its despeckling. Arch Comput Methods Eng. 2021;28:4633–53. https://doi.org/10.1007/S11831-021-09548-Z/TABLES/5. https://link.springer.com/article/10.1007/s11831-021-09548-z.
26. Solbø S, Eltoft T. Homomorphic wavelet-based statistical despeckling of SAR images. IEEE Trans Geosci Remote Sens. 2004;42:711–21. https://doi.org/10.1109/TGRS.2003.821885.
27. Touzi R, Vachon PW. RCM polarimetric SAR for enhanced ship detection and classification. Can J Remote Sens. 2015;41(5):473–84. https://doi.org/10.1080/07038992.2015.1110010.
28. Vitale S, Aghababaei H, Ferraioli G, et al. A multi-objective approach for multi-channel SAR despeckling. In: International Geoscience and Remote Sensing Symposium (IGARSS) 2021a, 2021-July, p. 419–22. https://doi.org/10.1109/IGARSS47720.2021.9553262.
29. Vitale S, Ferraioli G, Pascazio V. Multi-objective CNN-based algorithm for SAR despeckling. IEEE Trans Geosci Remote Sens. 2021;59:9336–49. https://doi.org/10.1109/TGRS.2020.3034852.
30. Wang P, Zhang H, Patel VM. SAR image despeckling using a convolutional neural network. IEEE Signal Process Lett. 2017;24:1763–7. https://doi.org/10.1109/LSP.2017.2758203.
31. Wang P, Zhang H, Patel VM. Generative adversarial network-based restoration of speckled SAR images. In: 2017 IEEE 7th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing, CAMSAP 2017, 2018, 2017-December, p. 1–5. https://doi.org/10.1109/CAMSAP.2017.8313133.
32. Yang Y, Newsam S. Bag-of-visual-words and spatial extensions for land-use classification. In: GIS: Proceedings of the ACM International Symposium on Advances in Geographic Information Systems, 2010, p. 270–9. https://doi.org/10.1145/1869790.1869829.
33. Yuan Y, Jiang Y, Wu Y, et al. Self-calibrated convolutional neural network for SAR image despeckling. In: International Geoscience

and Remote Sensing Symposium (IGARSS), 2021, p. 399–402. https://doi.org/10.1109/IGARSS47720.2021.9554769.
34. Yuan Y, Guan J, Feng P, et al. A practical solution for SAR despeckling with adversarial learning generated speckled-to-speckled images. IEEE Geosci Remote Sens Lett. 2022. https://doi.org/10.1109/LGRS.2020.3034470.
35. Yue DX, Xu F, Jin YQ. SAR despeckling neural network with logarithmic convolutional product model. 2018;39:7483–7505. https://doi.org/10.1080/01431161.2018.1471539. https://www.tandfonline.com/doi/abs/10.1080/01431161.2018.1471539.
36. Zhang Q, Yuan Q, Li J, et al. Learning a dilated residual network for SAR image despeckling. Remote Sens. 2018;10:1–18. https://doi.org/10.3390/RS10020196.

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